Lecture 11

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Midterm 1

- Reminder that Midterm 1 will be on Wednesday, October 14 in class.
- Midterm 1 will cover:
  - Chapters 1-5 and 8-12 in the text, and
  - the material in Lectures 1-8.
- The R material covered in lab will not be on Midterm 1 but is fair game for Midterm 2 and the final.

Midterm 1 (continued)

- You can find example questions for the exam from:
  - Sample midterms posted on Canvas
  - Homework questions
  - Additional questions in the text
  - Review questions
  - Questions discussed in lecture
- You will want to focus more on solving problems and less on memorizing the lectures.

Lecture 11

1. Example 1
2. The Law of Total Probability
3. Bayes' Rule
4. Example 2
5. Odds
6. Bayes' Rule in Terms of Odds
7. Example 2 in Terms of Odds
8. Discussion
9. Reading Summary - Chapter 15
10. Orderings, Permutations, and Combinations

Finish reading Chapter 15, start reading Chapters 16 and 17 for next time.

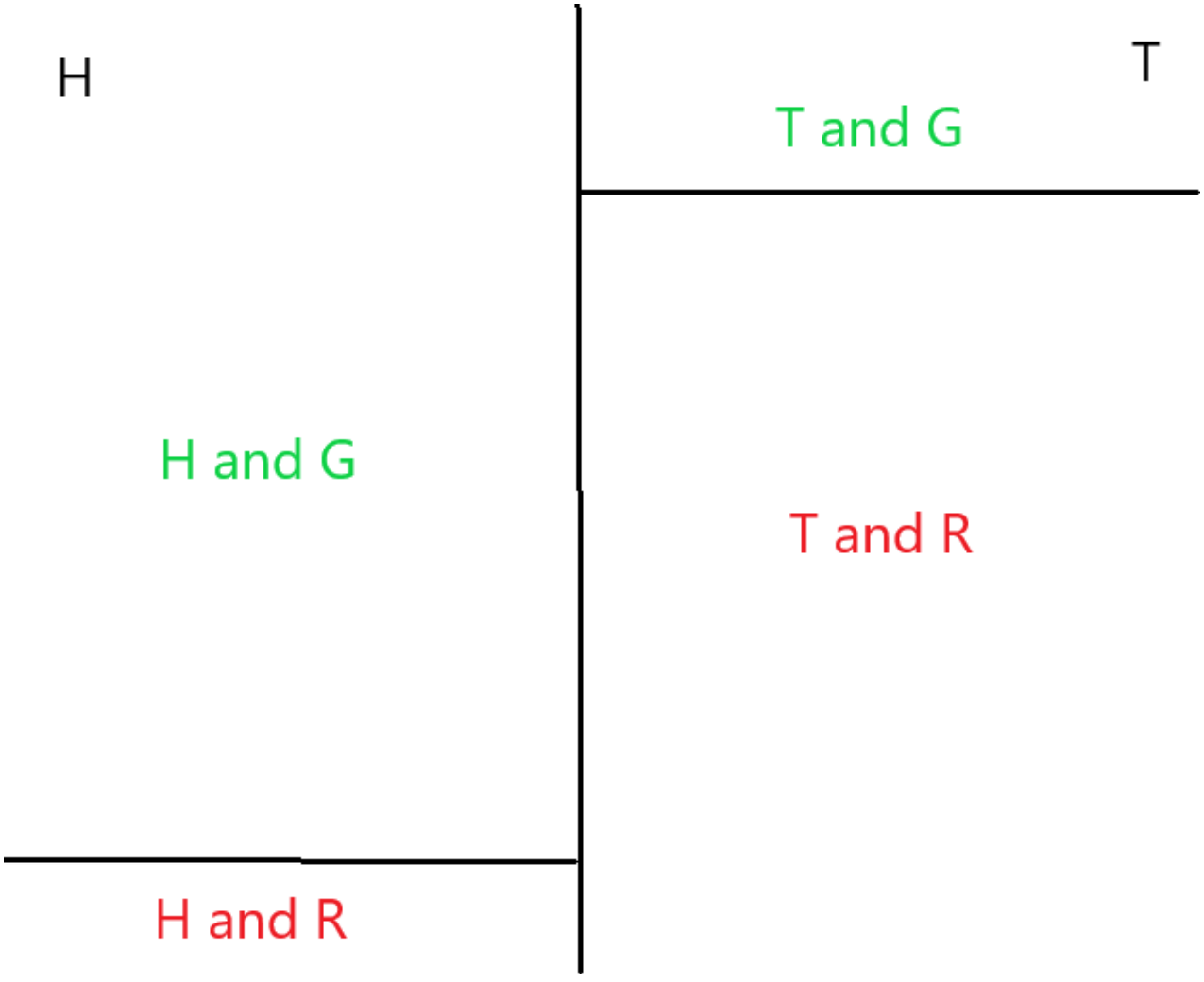
Example 1

- In jar A there is 1 red marble and 9 green marbles.
- In jar B there are 8 red marbles and 2 green marbles.
- You flip a coin.
  - If it lands heads, you pick a marble at random out of jar A.
  - If it lands tails, you pick a marble at random out of jar B.

Example 1 (Continued)

- What is the chance the coin lands heads?
  - We'll call this $P(H)$
- Given the coin lands heads, what is the chance of picking a red marble?
  - We'll call this $P(R|H)$.
- Given that the coin lands tails, what is the chance of picking a red marble?
  - We'll call this $P(R|T)$.
- What is the unconditional chance of picking a red marble?
  - We'll call this $P(R)$.

Venn Diagram



Law of Total Probability

- Note that $R \text{ and } H$ and $R \text{ and } T$ are mutually exclusive, and together they comprise R.
- Hence $P(R) = P(R \text{ and } H) + P(R \text{ and } T)$.
- We can now use the definition of conditional probability to rewrite this as:
  - $P(R) = P(R|H)P(H) + P(R|T)P(T)$.
- This is the **Law of Total Probability**.
- We can now calculate $P(R) = 0.1 * 0.5 + 0.8 * 0.5 = 0.45$.

Example 1 (Continued)

- Given you pick a red marble, what is the chance the coin landed heads? We'll call this $P(H|R)$.
- The last question is more challenging.
- Do you think that the chance is:
  - (i) about 50%,
  - (ii) much less than 50%, or
  - (iii) much more than 50%?
- Why?

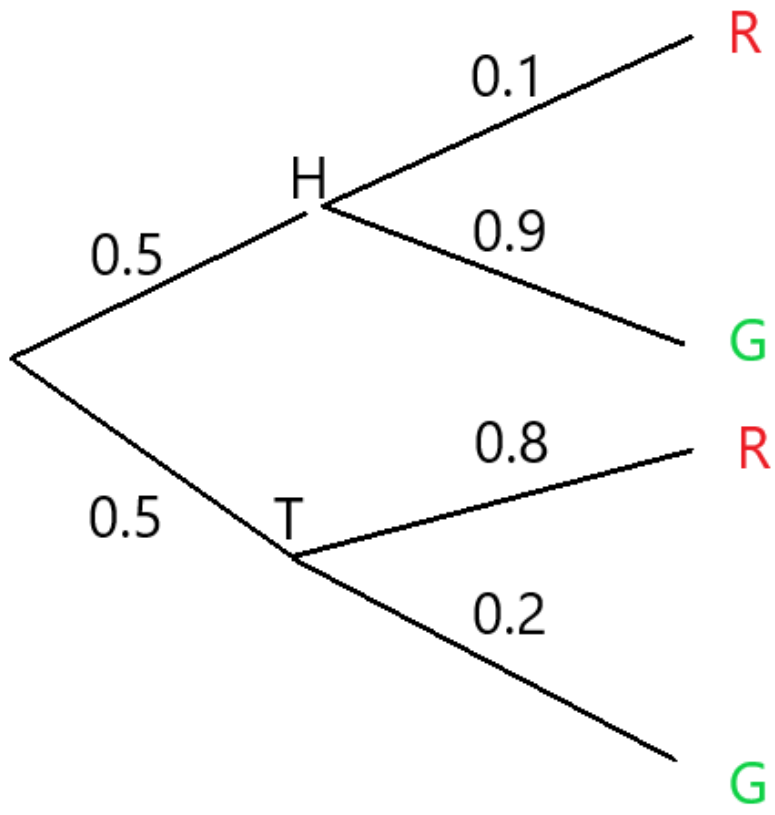
Bayes' Rule

- By the definition of conditional probability, we know that for two events A and B:
  - $P(A|B) = P(A \text{ and } B)/P(B)$.
- We also know from the definition of conditional probability that:
  - $P(A \text{ and } B) = P(B \text{ and } A) = P(B|A) * P(A)$.
- Hence $P(A|B) = P(B|A) * P(A)/P(B)$
- This is **Bayes' Rule**

Bayes' Rule (continued)

- Bayes' Rule allows us to calculate a conditional probability by reversing the direction of conditionality.
  - One direction is frequently more intuitive than the other.
  - The design of Example 1 makes $P(R|H)$ more intuitive than $P(H|R)$.
- We can now calculate $P(H|R) = P(R|H) * P(H)/P(R)$.
- This gives $P(H|R) = 0.1 * 0.5/0.45 = 0.11$.

Bayes' Rule in a Tree Diagram



Bayes' Rule in a Tree Diagram (Continued)

- To find the probability of a path, you multiply the probabilities along the path.
- To find the probability of an event, you add the probabilities of the paths that lead to that event.
- With this method, $P(H|R) = P(H \text{ and } R)/P(R)$
  - There is one path that leads to H and R and there are two paths that lead to R.
  - Hence $P(H|R) = 0.5 * 0.1/(0.5 * 0.1 + 0.5 * 0.8) = 0.11$.

Example 2

- A large company randomly tests its employees for drug use, and fires anyone who tests positive.
- As a result of this policy, only one out of 1000 employees use drugs.
- The test the company uses is not perfect, but if an employee is using drugs, there is a 98% chance of testing positive.
- If an employee is drug-free, there is a 99% chance of testing negative.
- If a randomly chosen employee tests positive for drugs, what is the chance the employee actually uses drugs?

Odds

- If $P(A)$ is the probability of some event A, then the **odds** of A are $P(A)/(1 - P(A))$.
- We can write the odds as $P(A)/P(\sim A)$, where $\sim A$ is the opposite event of A.
- What are the odds of A if $P(A)$ is:
  1. 0?
  2. 0.0001?
  3. 0.25?
  4. 0.5?
  5. 0.8?
  6. 0.99?

Odds (Continued)

- If the probability is small, then the odds and the probability are very close.
- If the probability is 0.5, then the odds are 1.
- The odds tend to ∞ as the probability tends to 1.
- If the odds are o and the probability p, then $p = o/(1 + o)$.
- Gamblers generally refer to the odds as opposed to the probability.

Bayes' Rule in Terms of Odds

- We can use Bayes' Rule to write the odds of $A|B$, referred to as the **posterior odds**, as:

$$P(A|B)/P(\sim A|B) = \frac{P(B|A) * P(A)/P(B)}{P(B|\sim A) * P(\sim A)/P(B)}$$

.

- We can then rearrange the right hand side of this equation as:
  - $(P(A)/P(\sim A)) * (P(B|A)/P(B|\sim A))$.

Bayes' Rule in Terms of Odds (Part 2)

- We call the first term $P(A)/P(\sim A)$ the **prior odds**.
  - This reflects the odds of A before taking into account any other information.
- We call the second term $P(B|A)/P(B|\sim A)$ the **data odds**.
  - This reflects the influence of new data B.

Bayes' Rule in Terms of Odds (Part 3)

- We can now restate Bayes' Rule as:
  $$\text{posterior odds} = \text{prior odds} * \text{data odds}\ .$$
- This gives statisticians a systematic way to update their expectations about the world when new data comes in.
- This is the basis of a branch of statistics called **Bayesian Statistics**
- **Frequentist Statistics**, the approach taken in this course, is the more traditional data analysis approach.

Example 2 Expressed in Terms of Odds

- Prior odds: $(1/1000)/(999/1000) = 1/999$
- Data odds: $0.98/0.01 = 98$
- Posterior odds: $98/999$.
- Probability: $98/(98 + 999) = 98/1,097$.

Discussion

1. Would you describe the drug test as accurate? Explain.
2. If you were a manager at this company, what might you suggest to do differently?

Reading Summary - Chapter 15

- Section 1: Introduction
  - An example of a binomial problem, step-by-step
- Section 2: The Binomial Formula
  - The general formula for binomial problems
  - Examples when the formula does not apply

Orderings, Permutations, and Combinations

Ordering a Set

- Suppose we have a set $\{a_1, a_2, a_3, ..., a_n\}$ of n objects.
- How many different ways can we order the n objects?
- Suppose we have a set $\{a_1, a_2, a_3\}$ of 3 objects (so $n = 3$).
- How many different ways can we order the 3 objects?

Ordering a Set with 3 Objects

- Answer: 6 :
  - (a_1, a_2, a_3),
  - (a_1, a_3, a_2),
  - (a_2, a_1, a_3),
  - (a_2, a_3, a_1),
  - (a_3, a_1, a_2),
  - (a_3, a_2, a_1).

Ordering a Set with 4 Objects

- Another example: suppose we have a set $\{a_1, a_2, a_3, a_4\}$ of 4 objects (so $n = 4$).
- How many different ways can we order the 4 objects?
- Answer: 24
- The 24 different ways are:
  - (a_1, a_2, a_3, a_4), (a_1, a_2, a_4, a_3),
  - (a_1, a_3, a_2, a_4), (a_1, a_3, a_4, a_2),
  - (a_1, a_4, a_2, a_3), (a_1, a_4, a_3, a_2),
  - (a_2, a_1, a_3, a_4), (a_2, a_1, a_4, a_3),

Ordering a Set with 4 Objects (Continued)

- (a_2, a_3, a_1, a_4), (a_2, a_3, a_4, a_1),(a_2, a_4, a_1, a_3), (a_2, a_4, a_3, a_1),
- (a_3, a_2, a_1, a_4), (a_3, a_2, a_4, a_1), (a_3, a_1, a_2, a_4), (a_3, a_1, a_4, a_2),
- (a_3, a_4, a_2, a_1), (a_3, a_4, a_1, a_2), (a_4, a_2, a_3, a_1), (a_4, a_2, a_1, a_3),
- (a_4, a_3, a_2, a_1), (a_4, a_3, a_1, a_2), (a_4, a_1, a_2, a_3), (a_4, a_1, a_3, a_2).

The General Rule

- For any n, the answer is:
  - $n! = n \times (n-1) \times (n-2) \times ... \times 2 \times 1$.
  - We read the value $n!$ as "n factorial."
- For example,
  - $3! = 3 \times 2 \times 1 = 6$, and
  - $4! = 4 \times 3 \times 2 \times 1 = 24$

Justification for the General Rule

- There are n options for the first object.
- There are then $n-1$ remaining options for the second object.
- There are then $n-2$ remaining options for the third object.
- ...
- There is then 1 remaining option for the last object.
- So there are a total of $n \times (n-1) \times (n-2) \times ... \times 2 \times 1$ possibilities.

Permutations

- How many ways are there to select and k objects from the previous set of n objects order matters?
- For example, suppose $n = 5$ and $k = 2$.

We have a set of 5 objects:

$$\{a_1, a_2, a_3, a_4, a_5\}$$

How many different ways can we select 2 objects from 5 where order matters?
- These are called **permutations**.

Permutations of 2 Objects from 5

- Answer: 20
  - (a_1, a_2), (a_2, a_1), (a_1, a_3), (a_3, a_1),
  - (a_1, a_4), (a_4, a_1), (a_1, a_5), (a_5, a_1),
  - (a_2, a_3), (a_3, a_2), (a_2, a_4), (a_4, a_2),
  - (a_2, a_5), (a_5, a_2), (a_3, a_4), (a_4, a_3),
  - (a_3, a_5), (a_5, a_3), (a_4, a_5), (a_5, a_4).

General Rule for Permutations

- For any n and any k, the answer is:
  - $\frac{n!}{(n-k)!} = n \times (n-1) \times (n-2) \times ...(n-k+1)$.
- For example,
  - $\frac{5!}{(5-2)!} = \frac{5!}{3!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1}$
  - $= 5 \times 4 = 20$.

Justification for the Permutations Formula

- Why does this work?
  - There are n options for the first object.
  - There are then $n-1$ remaining options for the second object.
  - There are then $n-2$ remaining options for the third object.
  - ...
  - There are then $n-k+1$ remaining options for the k-th object.
- This gives $n \times (n-1) \times (n-2) \times ... \times (n-k+1)$ possibilities.

Combinations

- How many ways are there to select k objects from n objects where we do not care about order?
- How many ways are there, for example, to select 2 objects from 5 objects?
- Answer: 10
  - $\{a_1, a_2\}$, $\{a_1, a_3\}$, $\{a_1, a_4\}$, $\{a_1, a_5\}$, $\{a_2, a_3\}$,
  - $\{a_2, a_4\}$, $\{a_2, a_5\}$, $\{a_3, a_4\}$, $\{a_3, a_5\}$, $\{a_4, a_5\}$,
- These are **combinations** or subsets.

Difference between Permutations and Combinations

- Note that
  - $(a_1, a_2) \neq (a_2, a_1)$, but
  - $\{a_1, a_2\} = \{a_2, a_1\}$.
- Use parentheses for permutations, where order matters, and brackets for subsets, where order does not matter.

Combinations Formula

- In general, for any n and any k, the answer is $\binom{n}{k} = \frac{n!}{(n-k)! \times k!}$.
- We read $\binom{n}{k}$ as "n choose k."
- This the number of subsets of size k in a set of size n.
- From the previous example:
  - $\binom{5}{2} = \frac{5!}{(5-2)! \times 2!}$
  - $= \frac{5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1) \times 2!}$
  - $= \frac{5 \times 4}{2!} = 10$

Justification for Combinations Formula

- Why does this work?
- Basic idea: each combination has $k!$ associated permutations.
- For example, $\{a_2, a_5\}$ has (a_2, a_5) and (a_5, a_2).
- In general, $\binom{n}{k} = \frac{n!}{(n-k)! \times k!} = \frac{n!}{(n-k)!} \div k!$

Main Point

- Given a set of n objects, the number of subsets of size k is equal to $\binom{n}{k} = \frac{n!}{(n-k)! \times k!} = n * (n-1) * \cdots * (n-k+1)/k!$.
- There are $\binom{n}{k}$ ways to choose k objects from n objects.
- Thus $\text{combinations} = \text{permutations} \div k!$,
- Or alternatively $\text{permutations} = \text{combinations} \times k!$

Properties of Combinations

1. $\binom{n}{k} = \binom{n}{n-k}$
   - Choosing the subset is equivalent to choosing the opposite of the subset.
2. $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$
   - A k element subset either contains the first element or it does not.
3. $\sum_{k=0}^{n} \binom{n}{k} = 2^n$
   - A set of size n has 2^n subsets.

- You can use property 2 to create Pascal's Triangle.

Pascal's Triangle

See Pascal's Triangle

R Code for Factorials, Permutations, and Combinations

```r
factorial(5) #5!.
```

```
## [1] 120
```

```r
#Use exp(lfactorial(n)) for large n.
#Permutations of 2 objects from 5
factorial(5)/factorial(5-2)
```

```
## [1] 20
```

```r
choose(5,2) #Combinations of 2 objects from 5
```

```
## [1] 10
```

Review Questions

- See Review_questions11.pdf on Canvas